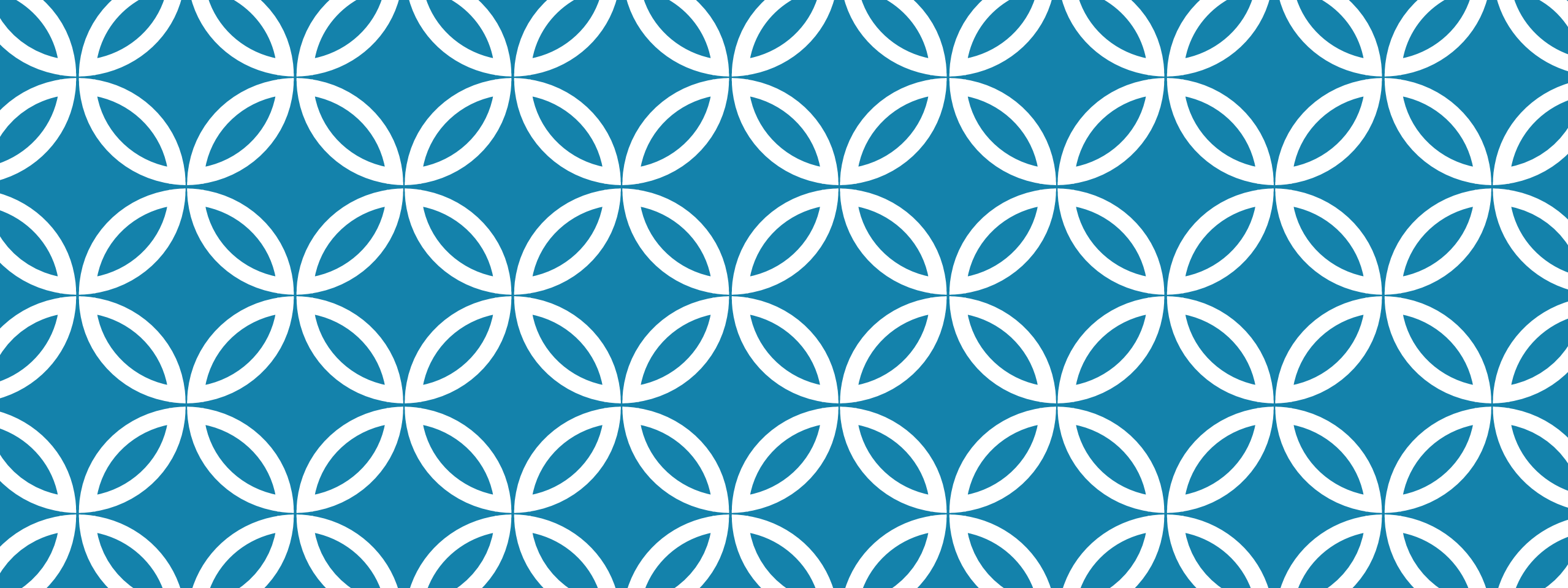




RESULT PRESENTATION & TECHNICAL WRITING STYLE

FIGURES, TABLES, AND EXPECTED RESULTS WRITING

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WHY THIS TOPIC MATTERS IN CSE

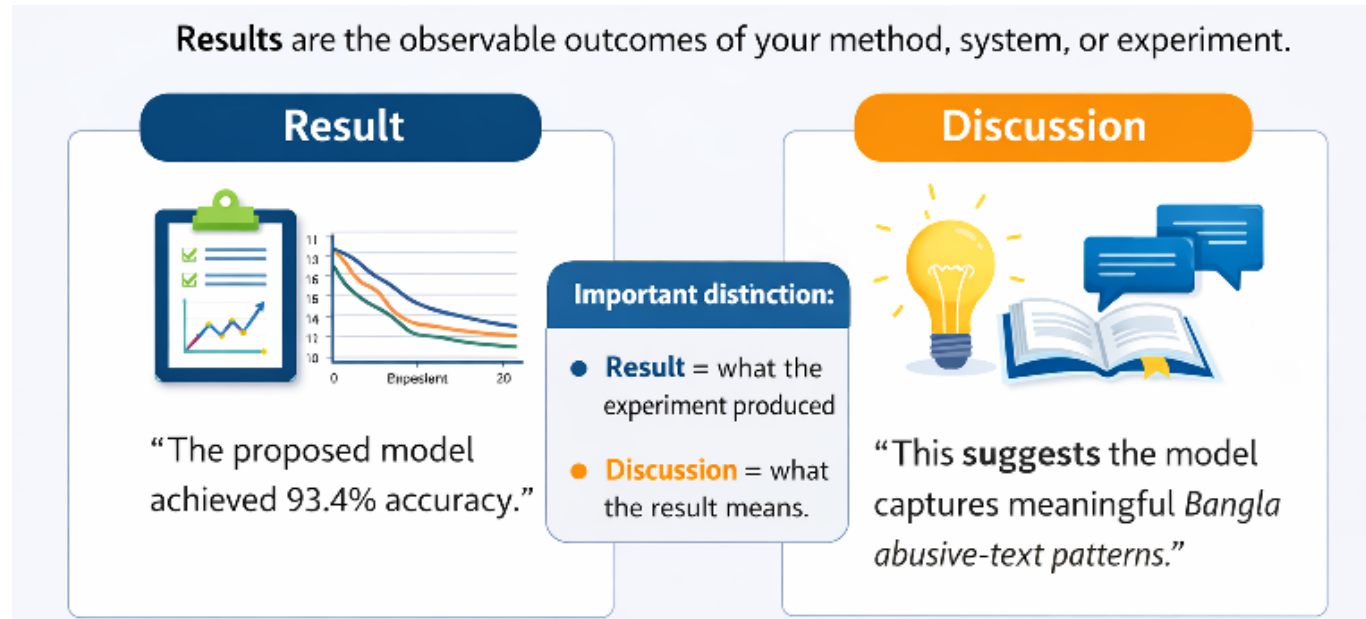
In CSE, research often produces:

- accuracy scores
 - runtime comparisons
 - confusion matrices
 - network performance metrics
-
- Readers do not only judge the method; they also judge:
 - clarity of presentation
 - correctness of interpretation
 - professionalism of writing
 - A weak presentation can make strong work look unreliable
 - A clear presentation helps in:
 - thesis writing
 - conference papers
 - project reports
 - final year design reports

WHAT COUNTS AS A “RESULT” IN CSE RESEARCH?

Common CSE result types:

- **Machine Learning:** accuracy, precision, recall, F1-score, ROC-AUC
- **Computer Networks:** throughput, latency, packet loss
- **Software Engineering:** defect counts, execution time, usability score
- **Databases:** query time, storage overhead
- **Cybersecurity:** detection rate, false positive rate
- **Image Processing:** PSNR, SSIM, classification accuracy



WHEN TO USE TEXT, TABLES, OR FIGURES

Use text when:

- only one or two values are important
- the result is simple and direct

Use a table when:

- exact values matter
- multiple systems or metrics must be compared
- the reader may want to look up specific numbers

Use a figure when:

- trends, patterns, or relationships are important
- changes over time or across models must be visualized
- you want to show comparison quickly

Rule of thumb:

- If precision matters, use a table.
- If pattern matters, use a figure.
- If the point is very small, use text.

CSE-based example:

- Accuracy of one model: write in text
- Accuracy, precision, recall, F1 for 5 models: use a table
- Accuracy trend across epochs: use a line graph

PRINCIPLES OF GOOD RESULT PRESENTATION

Good result presentation should be:

- **clear** — easy to read and interpret
- **accurate** — faithful to actual findings
- **relevant** — includes only useful information
- **consistent** — same labels, units, fonts, and terms throughout
- **reader-friendly** — understandable without verbal explanation

Analogy:

- *A result section is like a dashboard in a car.*
- *It should show the important information clearly, not overload the driver.*

Questions to ask before finalizing a result:

- What is the main message?
- Can the reader interpret it quickly?
- Are all labels, units, and terms clear?
- Is there unnecessary clutter?

DESIGNING EFFECTIVE FIGURES

A good figure should:

- have a clear purpose
- use readable axis labels
- include units where needed
- avoid unnecessary decoration
- use consistent scale and formatting
- be understandable in grayscale if printed

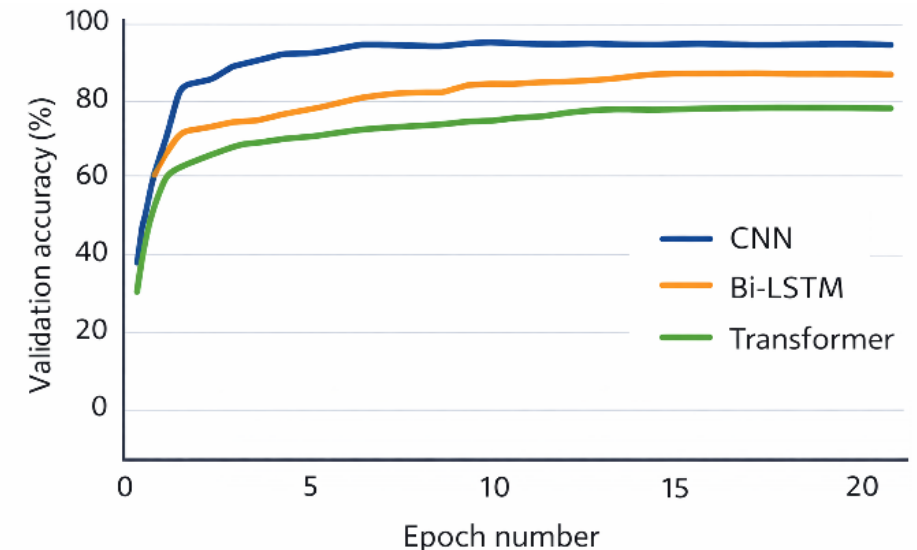
Best practices:

- label both axes properly
- start axes carefully; do not mislead with distorted scale
- use legends only when necessary
- keep text large enough to read
- avoid 3D charts unless essential
- highlight comparison, not artistic style

Example:

For model performance across epochs:

- X-axis: Epoch number
- Y-axis: Validation accuracy (%)
- Legend: CNN, Bi-LSTM, Transformer



COMMON FIGURE TYPES I

Bar chart

- best for comparing models, tools, or categories
- example: F1-score of 4 classifiers

Line graph

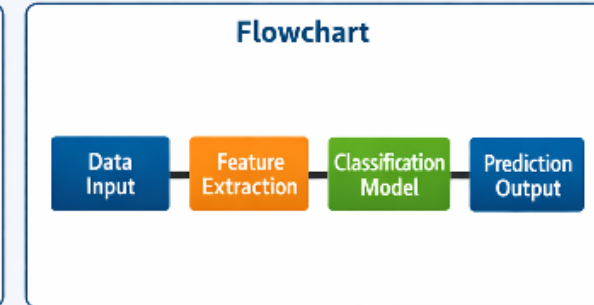
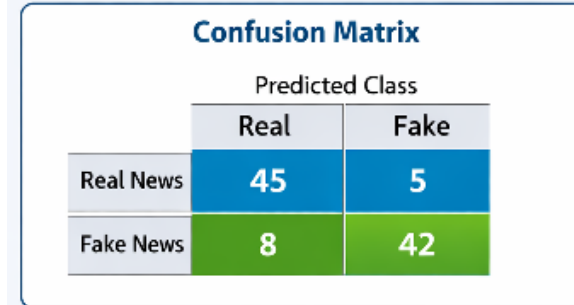
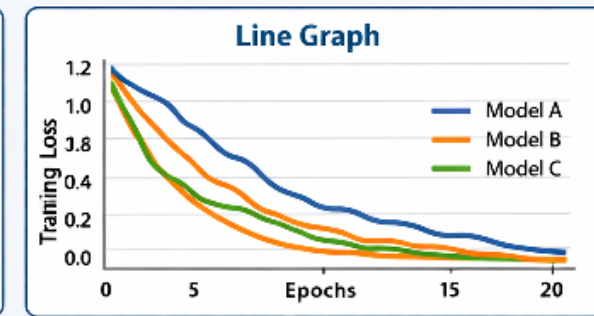
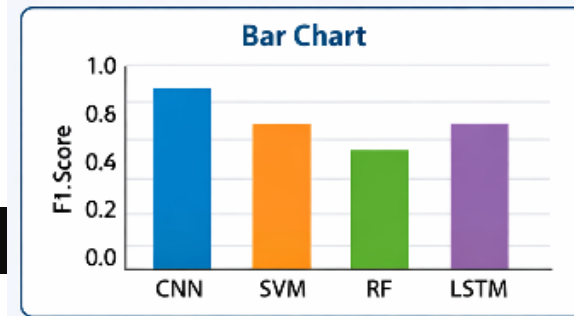
- best for trends over time or epochs
- example: training loss across iterations

Confusion matrix

- best for classification result distribution
- example: fake news vs real news predictions

Flowchart / architecture diagram

- best for explaining system design or workflow
- example: data preprocessing → feature extraction → classification



Scatter plot

- best for showing relationship between two variables
- example: response time vs server load

WRITING FIGURE CAPTIONS PROPERLY

A caption should tell the reader:

- what the figure shows
- what variables are being compared
- what context is important
- what abbreviations mean, if needed

Weak caption:

- “Model performance”

Better caption:

- “Figure 3. Comparison of F1-scores for CNN, Bi-LSTM, and SVM models on the Bangla abusive-text dataset.”

Caption writing tips:

- be specific, not vague
- mention dataset/task if relevant
- define unusual abbreviations

DESIGNING EFFECTIVE TABLES

Tables are useful when readers need exact values.

Best practices:

- use a short but informative title
- organize columns logically
- keep units visible in headers
- align decimal places
- avoid overcrowding
- include only necessary rows and columns
- highlight best value carefully if allowed by style guide

Do not:

- copy raw spreadsheet output directly
- use too many decimal places
- mix different units without explanation

LANGUAGE PATTERNS FOR WRITING RESULTS

To report a result:

- “The proposed model achieved...”
- “Table 3 shows that...”
- “Figure 4 illustrates the trend in...”

To compare results:

- “Model A outperformed Model B in...”
- “Compared to the baseline, the proposed method improved...”
- “A slight decrease was observed in...”

To describe trends:

- “Accuracy increased steadily across epochs.”
- “Latency decreased as cache size increased.”
- “The highest value was recorded for...”

To remain cautious:

- “The findings suggest that...”
- “This may indicate...”
- “A possible explanation is...”

Ex: “Figure 5 shows that validation loss decreased rapidly during the first 10 epochs and stabilized afterward.”

COMMON MISTAKES IN RESULT PRESENTATION

Title: Common Mistakes Students Should Avoid

Frequent mistakes:

- presenting raw output without interpretation
- repeating the same information in text, table, and figure unnecessarily
- using vague captions such as “Graph of results”
- including too many decimal places
- using inconsistent terminology
- claiming conclusions not supported by data
- confusing result and discussion
- writing expected results as if they are already proven facts

Poor sentence:

- “The model is perfect and works amazingly.”

Improved:

- “The model achieved the highest F1 -score among the compared methods under the current experimental setup.”

Case Study: Presenting Results for a Fake News Detection System

Scenario:

A student team builds a multimodal fake news detection system using text and image features.

Poor presentation:

- They write only: “Our model got 92% accuracy.”
- No baseline
- No table
- No class-level result
- No explanation of why the result matters

Improved presentation:

- **Table:** comparison with SVM, LSTM, and proposed model
- **Figure:** confusion matrix or bar chart of precision/recall/F1
- **Text explanation:** highlights strongest result and limitation
- **Expected result statement in proposal:** explains why multimodal input may improve classification

Key lesson:

- A result becomes convincing when it is structured, compared, and interpreted clearly.

IN-CLASS PRACTICE ACTIVITY

You are given the following raw result:

- CNN Accuracy = 91.2%
- Bi-LSTM Accuracy = 89.7%
- SVM Accuracy = 84.4%
- CNN F1 = 0.90
- Bi-LSTM F1 = 0.88
- SVM F1 = 0.82

Task:

- Decide whether to use text, table, or figure
- Write a suitable table title or figure caption
- Write 2–3 sentences describing the main finding

SUMMARY AND TAKEAWAYS

Results in CSE must be presented clearly, not just computed

- Use **text** for small points, **tables** for exact values, and **figures** for trends/patterns
- Good captions and titles are essential
- Expected results should be realistic and evidence-based
- Technical writing should be clear, formal, concise, and objective
- Strong presentation improves credibility, readability, and academic quality

Takeaway sentence:

- Good research is not only about finding results; it is also about communicating them professionally.

Suggested visual:

- Final summary mind map:
 - Results
 - Figures
 - Tables
 - Expected Results
 - Writing Style